# Hong Chul Nam

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## Education

**New York University** — New York City, USA
*Doctor of Philosophy in Data Science (CDS/Courant Institute)* — *Aug 2026 – Present*

**ETH Zurich** — Zurich, Switzerland
*Master of Science in Electrical Engineering and Information Science* — *Jan 2022 – Mar 2026*

**California Institute of Technology** — Pasadena, USA
*Visiting Researcher* — *Oct 2023 – Sept 2024*

**ETH Zurich** — Zurich, Switzerland
*Bachelor of Science in Electrical Engineering and Information Science* — *Sep 2018 – Sep 2023*

**St. Anselm's Abbey School** — Washington DC, USA
*High School* — *Sep 2014 - Sep 2016*

## Experience

**AI Researcher** — Apr 2026 – Present
*Zenithon AI* — *London, UK*

- One of AI researchers working on building a billion-parameter foundation model for plasma physics using transformer-based neural operators to accelerate physical simulation of nuclear fusion

**Visiting Student** — Jan 2026 – Present
*Harvard, Multiply Labs* — *Remote*

- Supervisor: Dr. Haonan Chen, Dr. Jiuguang Wang, Prof. Yilun Du
- Working on extending energy-based models into Lagrangian formulation for developing energy-based world models for video generation and policy learning

**AI Researcher** — Oct 2024 – Present
*Korea Military Academy* — *Seoul, Republic of Korea*

- Supervisor: Sunil Hwang, Prof. Hyun Kwon
- Mandatory military service (1.5 years)
- Participated (top 2 out of 270 teams) in the public competition for AI-based automatic document generation using VLM for in-home elder care work
- Built an autonomous surveillance system using real-time object detection with VLM for the Korean army
- Achievements: publication in CoRL Workshop 2025, ICML Workshop 2026, submission to NeurIPS 2026, second-place in competition for document automation for in-home elder care works

**Student Researcher** — Oct 2023 – Present
*Anima's Lab, California Institute of Technology* — *Pasadena, USA*

- Funding: Caltech Research Funding
- Supervisor: Dr. Julius Berner, Dr. Xi Deng, Prof. Anima Anandkumar
- Solved high-dimensional Poisson equations using stochastic reformulation via Walk-on-Spheres (WoS) and neural network in a general framework called Neural Walk-on-Spheres (NWoS)
- Extending self-supervised learning methods (VICReg, MAE, JEPA) to functions with neural operators
- Collaborating with researchers from NVIDIA, University of Chicago, Argonne National Laboratory and Caltech to improve weather modeling with self-supervised learning
- Collaborating with a researcher from Purdue University to extend Neural Walk-on-Spheres (NWoS) to neural operators (WoS-NO)
- Co-supervising a student on neural operator learning for path planning via screened Poisson equations and WoS.
- Generalizing random walk schemes on training data generated from expensive mesh-based solvers for learning transition kernel (i.e. Green's function) to enable generalization of stochastic solvers into nonlinear PDEs
- Achievements: publication in ICML 2024, submission to ICLR 2026

**AI Researcher** — Jan 2022 – Dec 2024
*Alsemy Inc.* — *Seoul, Republic of Korea*

- Supervisor: Chanwoo Park, Dr. Ye Sle Cha, Dr. Jongwook Jeon, Hyunbo Cho
- Accelerated modeling of variability in transistor devices under process variations using generative models and importance sampling
- Implemented neural operator-based foundation models and reinforcement learning methods for transistor device modeling and circuit optimization

- Created the first function-level anomaly detection neural operator for detecting anomaly device simulations
- Created the device simulation augmentation method by training a function-level autoencoder and latent diffusion model to improve the diversity of large-scale training data
- Proposed a neural ODE-based compact model for transient self-heating simulation of transistor devices
- Achievements: publications in SISPAD 2023/2024, NeurIPS 2023 Workshop, ICMC 2025, IEEE Electron Device Letters Submission; full simulation pipeline for variability analysis, DTCO, anomaly detection and self-heating

**Research Assistant** — Sep 2021 – Jul 2023
*SAFARI Lab, ETH Zurich* — *Zurich, Switzerland*

- Supervisor: Konstantinos Kanellopoulos, Prof. Onur Mutlu
- Created a neural network-based page table walk cost estimator to selectively place evicted L2 TLB page tables into L2 data cache
- Introduced Merkle trees in page tables to enforce efficient security checks during each intermediate walk of page table walks to verify the integrity of page tables under security attacks
- Contributed to CUDA programming introductory codes for a heterogeneous computing course
- Achievements: publication in MICRO 2023; extension of SniperSim to support neural network; extension of Gem5 and SniperSim to support Merkle trees in page tables

**Research Intern** — Sep 2021 – Apr 2022
*Swiss Center for Electronics and Microtechnology (CSEM)* — *Zurich, Switzerland*

- Supervisor: Dr. Andrea Bonetti
- Developed a low-power MFCC feature extractor using SystemVerilog with UVM verification for a speech signal AI accelerator
- Explored structural changes in hardware as well as neural network architecture for achieving ultra-low power consumption
- Achievements: implementation of MFCC module in SystemVerilog code with verification; optimization of hardware/software implementation of the block

**Research Engineer** — Sep 2020 – Sep 2021
*ASL Lab, ETH Zurich* — *Zurich, Switzerland*

- Funding: Swiss military, ETH Zurich
- Supervisor: David Rohr, Dr. Nicholas RJ Lawrance, Prof. Roland Siegwart
- Developed an autonomous firefighting VTOL drone for wildfire surveillance from scratch
- Achievements: development of VTOL drone with surveillance system

**Research Assistant** — Jun 2015 – Jul 2015
*Vanderver Laboratory, Children's National Hospital* — *Washington DC, USA*

- Funding: Summer internship stipend
- Supervisor: Dr. Asako Takanohashi, Prof. Adeline Vanderver
- Worked on Aicardi Goutieres Syndrome using PCR, cell culture, western blot, and immunohistochemistry approaches
- Achievements: learning cellular biology and all techniques for growing cells and treating drugs

## Projects

**Self-supervised Learning for Few-Step Models** — Jun 2025 – Present

- Collaborator: Sunil Hwang, Junseob Kim, Jaehyeong Jo
- Quantifying the loss of sample diversity in few-step models such as the consistency model and meanflow
- Introducing semantic information with alignment to improve the diversity

**Debiasing the Classifier-Free Guidance due to Nulltoken** — Dec 2024 – Present

- Collaborator: Joochul Lee
- Proved quantitatively the bias in probability distributions due to null token for approximating the unconditional distribution in conditional flow matching models and working on removing the bias

**Q-Guided Flow Q-Learning** — Dec 2024 – Sep 2025

- Supervisor: Prof. Hyun Kwon
- Developed a decoupled actor–critic framework combining flow-based policy learning and value-guided correction for stable and expressive offline reinforcement learning

**Continuous Control of Robot Arm** — Jan 2023 – Jun 2023

- Supervisor: Bhavya Sukhija, Lenart Treven, Prof. Andreas Krause
- Implemented robot API for RL in the real world for the UR5 robot

- Implemented and tested offline model based RL methods on the robot
- Achievements: implemented software interface to control UR5 robot arm; verified RL methods in the robot

**Autonomous Driving Mini-car** Sep 2020 – Jan 2021

- Supervisor: Dr. Michele Magno, Prof. Luca Benini
- Developed a real-time autonomous driving model using a Sony microcontroller and an RC car with external camera modules for CNN-based lane detection
- Achievements: demonstrated the highly accurate and functional autonomous driving system with Sony microcontrollers using CNN-base architectures

## Publications


- **Hong Chul Nam***, Jeonghwan Cheon*, Jaemo Shin, Hyun Kwon. "Energy-Based Operator Learning in Function Space". Submission: Conference on Neural Information Processing Systems (NeurIPS 2026)
- Hrishikesh Viswanath*, **Hong Chul Nam***, Julius Berner, Anima Anandkumar, Aniket Bera. "Operator Learning Using Weak Supervision from Walk-on-Spheres". In: arXiv preprint arXiv:2603.01193 (2026).
- Eunki Joung, **Hong Chul Nam**, Geonhwi Hwang, Moohyun Lee, Hyun Kwon. "Making Visible, Making Invisible: How an AI Scribe Reshapes Documentation Authority in Social Work". In: Trustworthy AI for Good (AI4GOOD) Workshop@ ICML 2026.
- Yejun Jang*, **Hong Chul Nam***, Jeong Min Park*, Gimin Bae, Hyun Kwon. "Q-Guided Flow Q-Learning". In: CoRL 2025 Workshop RemembeRL
- Premkumar Vincent*, Yeongwoo Nam*, Kyungmin Kim*, **Hong Chul Nam**, Hyunseok Whang, Donghyun Jin, Jongwook Jeon, Chang-Sub Lee, Jihun Park, Ye Sle Cha, Hyunbo Cho. "Large Pre-trained Model Approach for Efficient Design Technology Co-Optimization". **(Oral)** In: International Compact Modeling Conference 2025 (ICMC 2025).
- **Hong Chul Nam***, Tae Il Oh*, Ye Sle Cha, Jongwook Jeon, Hyunbo Cho. "FuncFlow: A Generative Neural Operator Using Diffusion Model for Simulation Augmentation". **(Oral)** In: International Compact Modeling Conference 2025 (ICMC 2025).
- **Hong Chul Nam***, Julius Berner*, Anima Anandkumar. "Solving Poisson Equations using Neural Walk-on-Spheres". In: Forty-first International Conference on Machine Learning (ICML 2024).
- Tae Il Oh*, **Hong Chul Nam***, Chanwoo Park, Hyunbo Cho. "FuncAnoDe: A Function Level Anomaly Detection in Device Simulation". **(Oral)** In: International Conference on Simulation of Semiconductor Processes and Devices (SISPAD 2024).
- Konstantinos Kanellopoulos, **Hong Chul Nam**, Nisa Bostanci, Rahul Bera, Mohammad Sadrosadati, Rakesh Kumar, Davide Basilio Bartolini, Onur Mutlu. "Victima: Drastically Increasing Address Translation Reach by Leveraging Underutilized Cache Resources". In: Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO 2023).
- Chanwoo Park*, **Hong Chul Nam***, Jihun Park, Jongwook Jeon. "FlowSim: An Invertible Generative Network for Efficient Statistical Analysis under Process Variations". **(Oral)** In: International Conference on Simulation of Semiconductor Processes and Devices (SISPAD 2023).
- **Hong Chul Nam**, Chanwoo Park. "NPC-NIS: Navigating Semiconductor Process Corners with Neural Importance Sampling". In: NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World.


## Awards

**Second place (top 2 / 270 teams)**: Competition for in-home elder social care (Aug 2025)
**Third place**: Defense Technology Startup Competition, Korean Army (May 2025)
**Fourth place**: Army Startup Competition (Mar 2025)

## Technical/Linguistic Skills

**Linguistics**: English, Mandarin, Korean, German
**Languages**: Java, Python, C/C++, Verilog/SystemVerilog
**ML Libraries**: PyTorch, TensorFlow, JAX
**Computer Simulators**: Gem5, SniperSim